Islamic University of Technology
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Project Odyssey

Software Development Project Proposal

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1. Project Motivation

We often find ourselves lost when we want to travel to a new location. Modern travel planning is currently a fragmented and overwhelming process. Whether it's for leisure or for some event, when ever we want to tour some location it's always not that easy to find all the "must-see" and "must-do" places. Often before going on the trip we have to go down a rabbit hole onTripAdvisor for reviews, YouTube for visual validation, and spreadsheets for budgeting etc. This leads the travelers to suffer from "information overload" and "analysis paralysis.". There is not one unified platform to research the trip, plan it and execute it properly. Our platform aims to fill that void on the internet by brining all the features one needs to make a memorable trip.

Traveling is a signifiant investment of time and money. So it is important that we avoid all kind of hindrances in our travels like inefficient routes, missed "hidden gems," budget overruns, and stress, ultimately diminishing the experience all because of a poor plan.

Context & Current Challenges: Existing platforms are not connected to each other. The review sites do not offer an effective route planning option, the map applications are not social enough and do not include itinerary planning, and social media cannot be useful in terms of logistics. Moreover, new travelers have a tendency of failing to make realistic schedules causing improbable schedules.

2. Project Objective

Our goal is to develop a unified Trip Planning and Sharing platforms that acts as a one-stop solution for travelers, a kind of s "Super App" for the outward and venturing:

Expected Outcomes:

- A platform where users can discover locations categorized by interest (Nature, Urban, History). A platform where users can discover tourist locations and interests categorized under three categories:

    A. Nature & Exploration

    B. Urban Lifestyle

    C. History and Museums

- An AI-assisted planning tool that generates optimized itineraries based on group size, duration, and preferences. Or the users can do it manually also using our intuitive interface.
- Create a community-driven database of verified trip plans and locations.
- A platform where users can clone and adapt itineraries from previous travelers.
- A digital "Memory Lane" that automatically chronicles a user's travel history thus allowing users to digitally preserve their travel memories in a timeline format.
- Automating itinerary creation using the planning tools.
- Provide travelers with optimized route plans to save time and money.
- Arranging groups for trips and activities that require a group to be present.
- Get price estimates on tickets, accommodation, commute etc
- Allow users to blog about their trip and how it went with reviews, suggestions etc.

3. Major Functional Requirements

1. Discovery & Categorization

- **Location Browser:** The users will be able to see the tourist locations, attractions and points of interest for any location they search for on the platform categorized into the following categories:

    B. Nature & Exploration

    C. Urban Lifestyle

    D. History & Museums.

- **Search & Nearby:** Users can search for specific places or use geolocation to find "Places to Visit" nearby.

2. Smart Planning Tools

- **AI Itinerary or planner:** An AI assistant will be able to help the users plan their journey and discover locations on their trip through natural conversation, making it a seamless experience. The AI will able able to suggest the users locations to visit and other Q&A based on the tastes of the users. Data privacy will be ensured.
- **Route Optimization:** The system will suggest the most efficient path (sequence of visits) based on the geographical location of attractions. This will suggest to visit the attractions in clustered locations making their journey smooth and removing repeated visits to a location which wastes a lot of time.
- **Price Estimation**: Provide estimated costs for the trip based on selected activities.
- **Manual Planning:** Users will also be able to manually make their plans using the robust UI present.

3. Social & Community

- **Trip Sharing:** After users make their itineraries they can share it if they. This will allow the system to suggest itineraries to future travelers with similar duration of stay and group size.
- **Group Finder:** A feature to connect solo travelers or small groups looking for companions. Specially for activities that cannot be participated solo or require a certain group size.

- **Content Contribution:** Users can upload photos and videos to specific location pages. They can also review or comment on places as to how to enjoy them to the fullest or even if there is anything they need to look out for!
- **Blogging:** Users can blog on their profiles on trips they have been to and share their experience. This also allows users to make the platform their own travel blogging space.

4. Personal Utilities

- **Memory Lane (Timeline):** A chronological timeline view showing a history of all visualization of all places a user has visited over time.
- **Offline Mode:** Ability to save itineraries and maps for use in areas with poor connectivity.

5. Future Expansion

In future, vendors may be allowed to sell packages on the platform for a smooth and easy experience for the travelers to engage in group packages early on instead of going to the location and then searching for travel agencies.

4. Breakdown of the Problem

We have divided the system into the following modules to face the complexity of a unified travel platform. The modules are described below:

1. **User Management Module:** This module helps in handling security, identity, and the feature of "Memory Lane".
   - **Authentication & Role Management:** Our system will differentiate among Travelers (consumers), Vendors (service providers), and Admins. It will manage login system (OAuth) and session management.
   - **Preference Profiling:** Our system will collect Travel DNA, which refers to gathering data on group size, chosen pace (comparison between relaxed and fast), and interests (Nature, Urban, History) . This information will be provided to the recommendation engine later.
   - **Digital Memory Bank:** Our platform will convert an unstructured user data into a structured chronological timeline, which would require efficient storage and proper retrieval of media assets.
2. **Data Acquisition Module:** This module will play a vital role in handling the database of locations, categorizations (Nature, Urban, Historical places, etc.), and media which will be created by user (photos/videos).
3. **Optimization Engine:** We will use the algorithmic core that manages the logic of "Traveling Salesman", which means to calculate the best path and produce AI routes that will be based on constraints (time/budget).
4. **Social Interaction Module:** Our system will have this module to govern the sharing of itineraries, blog posts, comments, and the logic of Group Finder.
5. **Vendor/Business Module (Future Expansion):** This module allows third-party sellers to enter their packages. Moreover, it will also manage the display of these type of offers to users. For now, we are keeping it for future expansion.

5. Timeline / Work Plan

1. Major Tasks

The work has been broken down into technical setup, core feature development and final polish to accomplish the project objectives.

- **Infrastructure & Design:**
  A. Create GitHub repository and configure MERN project structure.
  B. Create low-fidelity wireframes on Figma and establish the MongoDB Schema.
  C. Add Cloudinary to work with media and Redux Persist to have offline functions.
- **Core Development (MVP):**
  A. Implement User Authentication (Sign-up/Login) using JWT or Clerk.
  B. Integrate Mapbox or Google Maps APIs to visualize the place.
  C. Develop CRUD functionality Itinerary Builder and search logic attractions.
- **Advanced Functionality & AI:**
  A. Create the Discovery social feed and public / private trip switch.
  B. Combine the OpenAI API and architect prompts to translate textual input of the users into JSON itineraries.
  C. Implement route optimization logic.
- **Deployment & Quality Assurance:**
  A. Conduct end-to-end testing and bug fixing.
  B. Apply UI/UX polish using Framer Motion.
  C. Deploy the frontend to Vercel and the backend to Render or Railway.

2. Milestones

These milestones indicate the achievement of certain Focus Areas upon the termination of every Sprint cycle.

Milestone 1: Inception & Architecture Ready –Finalize structure, database schema and the design assets of the project are completed.

Milestone 2: MVP Foundation – Customers are able to log in, and the system is able to store and display simple locations on a map.

Milestone 3: Functional Itinerary Builder – The users are capable of creating, maintaining, and visualizing trip routes.

Milestone 4: Social Connectivity – The application will encourage social connection with your community in terms of feed sharing and photo-posting.

Milestone 5: AI-Powered Automation – The system is able to auto-generates trips through the use of natural language prompts.

Milestone 6: Feature Complete – The app is optimized and the offline modes and group features are functional.

Milestone 7: Live Deployment – - The application is refined, tested and deployed on live production servers.

3. Estimated Completion Dates

The estimates of each sprints have been displayed below.

Sprint Durations: 2 weeks. No of Sprints: 6-7

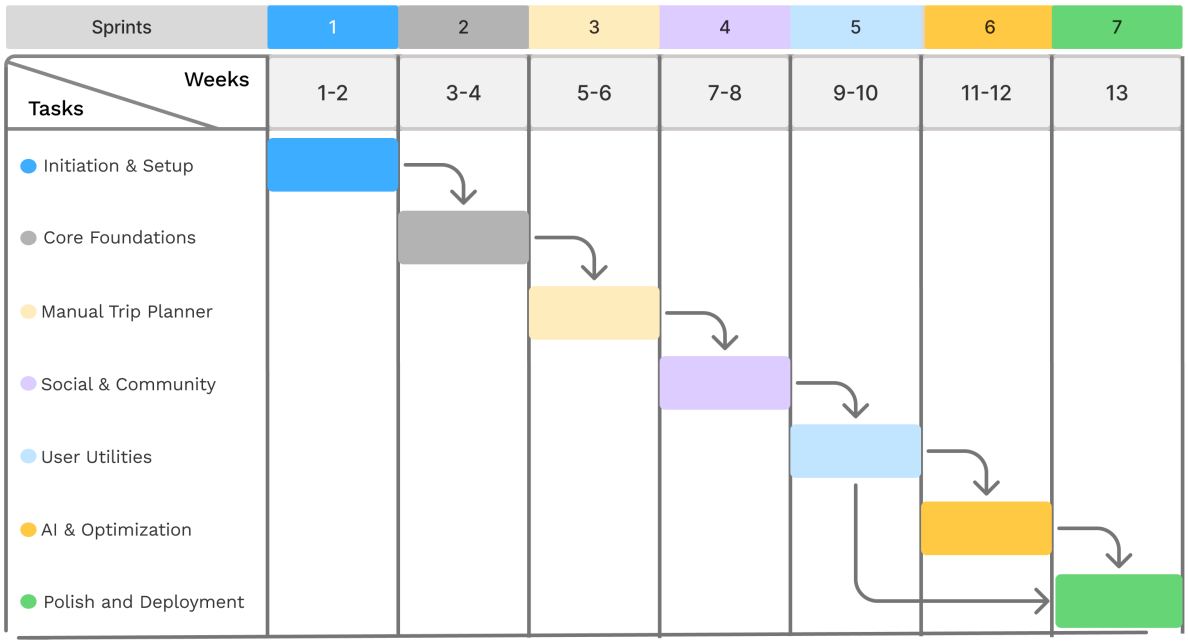

Fig-01: Timeline for Project Odyssey

6. Major Stakeholders

1. End Users (Primary Users)

- Anyone planning trips.
- Plan trips, save places, discover destinations, and share experiences.
- Provide feedback and suggestions.

2. Admins / Platform Managers

- Staffs responsible for managing the platform.

3. Developers / Technical Team

- Software engineers.
- Build, test, and deploy the system.

4. Investors / Project Sponsors

- Individuals or organizations funding the project.

5. Community / Social Groups

- Travel bloggers, local guides, tourism boards, or user communities.
- Contribute content, reviews, tips, and recommendations.

7. Hardware / Software / Tools Requirements

The tools and technologies that we are going to use throughout our project have been listed below:

1. Programming Languages

- JavaScript / TypeScript : We will use it for frontend and backend development.
- HTML / CSS: We will use it for UI structure and styling.
- SQL : We will use it for database queries.
- Java or Kotlin: To build an android app, we are going to use Java/Kotlin.
- Swift: We will use Swift for future iOS app.

2. Frameworks and Third-party Libraries

Frontend:

- React.js
- Next.js: We will use it for server-side rendering and routing.
- Tailwind CSS: We will use it for styling and responsive UI.
- Leaflet: We will use it for creating interactive maps and optional Google Maps API if needed.

Backend:

- Node.js with Express
- JWT / OAuth libraries (for authentication)
- WebSocket libraries (for real-time updates, optional)

Maps and GIS:

- OpenStreetMap (base map data; optional Google Maps)
- PostGIS (spatial functions for PostgreSQL)

Storage:

- Cloudinary or AWS S3 SDK (for media storage)

Other utilities:

- Axios or Fetch API (for HTTP requests)
- Lodash or other utility libraries as needed

3. Hardware Components (if any)

Developer machines (laptops/desktops) with:

- Minimum 8 GB RAM, recommended 16 GB
- Modern CPU (i5/Ryzen 5 or higher)
- SSD storage for faster builds
- Internet connectivity for cloud services, map tiles, and API usage
- Mobile devices for testing responsiveness and mobile UI

4. Databases

- Primary: PostgreSQL with PostGIS (for map-heavy operations)
- MongoDB (for user-generated content and flexible document storage)

5. Testing Tools

- Unit Testing: Jest / Mocha (Node)
- End-to-End Testing: Cypress
- API Testing: Postman

6. Bug Management and Configuration Tools

- Version Control: Git + GitHub (branching, merging, CI/CD)
- Issue Tracking / Bug Management: GitHub Issues or Jira
- UI/UX Prototyping: Figma
- Build Tools & Package Management: npm / yarn
- Deployment & Configuration: Docker (optional), Netlify / Vercel (frontend), Heroku / AWS (backend)

8. Critical Challenges of the Project

Developing a comprehensive Trip Sharing and Planning Platform involves several technical, operational, and resource-related challenges. The major anticipated difficulties are:

1. Technical Complexity

Developing a smart trip-planning platform involves several components that require acute optimization and design.

- **Real-Time Geolocation & Mapping**

Integrating mapping functionality involves handling live location data, map layers, coordinate systems, and nearby-search algorithms. In addition, maintaining smooth map rendering and interactions, particularly on mid-range or low-end devices requires careful optimization and efficient use of resources.

- **Route Optimization Algorithms**

Creating an efficient route planner is far from trivial. It involves implementing advanced algorithms such as shortest-path methods, the Traveling Salesman Problem (TSP), or various heuristic and meta-heuristic approaches.

The real challenge lies in striking the right balance between accuracy and performance, especially when users add multiple stops or request highly customized routes. Achieving fast, reliable results at large scale requires significant algorithmic tuning.

- **AI-Based Itinerary Generation:**

It requires training or designing models capable of understanding preferences, group size, duration, and location data. Ensuring the generated itinerary is realistic and personalized adds extra complexity.

2. Data Availability

A major challenge in building the platform is securing reliable, comprehensive, and consistently updated data.

- **Accurate Location Data:**

Essential information such as attraction coordinates, opening hours, entry fees, and detailed descriptions is often scattered across multiple sources. No single dataset provides everything, which means some information must be manually curated or assembled from various public APIs. Maintaining the accuracy and freshness of this data adds ongoing operational effort.

- **Community-Generated Content:**

The platform relies heavily on users to upload photos, videos, and reviews to create a richer and more engaging experience.However, this introduces challenges such as verifying content quality, preventing misinformation, and avoiding duplicate or irrelevant submissions. Establishing moderation guidelines and automated validation tools becomes important to keep the platform clean and trustworthy.

3. Resource Constraints:

Some features require paid APIs (e.g., Google Maps, Places API). Free tiers have limitations on daily usage, requests, or map loading. So budget can become an issue. Storage requirement is also a challenge as hosting large amounts of media (photos, videos) demands significant storage and bandwidth. Cloud storage may be necessary, adding cost and complexity.

4. Integration Issues

- **API Integration:**

Integrating mapping APIs, routing services, image uploads, and AI micro-services may cause compatibility issues. APIs may have rate limits or versioning changes.

- **Offline Functionality:**

Supporting offline usage such as accessing saved trips when users have limited connectivity requires additional layers of caching, local storage, and synchronization

logic. Keeping offline itineraries updated with their online versions demands careful conflict resolution and version control.

9. Conflicting Requirements

During the planning of this Trip Sharing and Planning Platform, several requirements may conflict with each other or require design trade-offs. Some requirements also need further clarification or negotiation to reach a balanced and feasible solution.

1. Performance vs. Accuracy in Route Optimization

Conflict: Highly accurate routes (like TSP-based solutions) need heavy computation, which slows down performance, especially on mobile devices.

Trade-off: Use faster heuristic or approximate algorithms.

Needs Clarification: Should the system prioritize shortest time, shortest distance, or user preference?

2. Personalization vs. Privacy

Conflict: Better recommendations require collecting user data, but too much data raises privacy concerns.

Trade-off: Collect minimal data at the cost of less personalization.

Needs Clarification: What data are users comfortable sharing?

3. User Freedom vs. Recommendation Accuracy

Conflict: Allowing users full freedom to create any itinerary may lead to illogical routes. Restricting them improves optimization, but reduces usability.

Trade-off: Let users override AI suggestions at the cost of efficiency.

Clarification Needed: Should the system enforce optimal routes or simply suggest them?

4. Open Contribution vs. Data Consistency

Conflict: Users can add missing places, but inconsistent naming, descriptions, or categories reduce data quality.

Trade-off: Limit contributions to predefined categories or enforce strict templates.

Clarification Needed: Should users freely add categories, or must they choose existing ones?

5. UI Simplicity vs. Feature Density

Conflict: A clean, simple UI is ideal, but the platform has many tools (maps, itinerary planner, social feed, blogs, vendor section), which may clutter the interface.

Trade-off: Using layered navigation or modular design but adding extra steps for the user.

Requires Clarification: What is the primary user flow we want to prioritize?

These conflicting requirements highlight the need for balancing user expectations, technical feasibility, and project constraints. Many of these will require team discussion and stakeholder negotiation to finalize a stable set of requirements for development.

— End of Report —